Prions

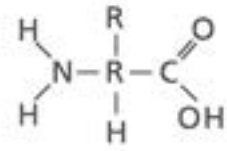
Prions: mysterious 'brain eating' proteins

When we think of disease-causing agents, most people think of bacteria, viruses, fungi or parasites. While these are often the cause of disease, the *prion* (pronounced *pree – on*) now presents a challenging alternative. The existence of disease-causing prions, which infect the brain, is a hotly debated issue. However, evidence suggests that they are real and that they do cause disease. The question is *how*?

What is a prion?

Prions are proteins that are made of about 250 amino acid groups. They are produced in the cell and are a part of our bodies. Not all prions behave in a 'natural', healthy way. Rogue prions have been identified. They have a recognisably deformed structure and are believed to cause the 'prion diseases'. The deformed prions, of which about 20 variations have been recognised to date, only differ from healthy ones in the way they are folded. The different shape appears to be linked to their ability to cause disease. (Imagine creating an animal shape using origami then, using exactly the same sheet of paper, opening it out and refolding it into another shape.)

Structure of an amino acid



(R = organic group)

1 Name the elements present in an amino acid.

Prion diseases

In 1986, England experienced its first 'Mad cow' epidemic. Mad cow, or Bovine spongiform encephalopathy – BSE, is an incurable disease that causes parts of the brain to be eaten away. This leads to coma and death within about 3 to 12 months of infection. The name for BSE (pronounced bo – vine sponj – ee – form – en – sef – alo – path – ee) is derived thus:

- *bovine:* refers to cows; *spongiform*: sponge-like; *encephalopathy:* brain degeneration.

The brain tissue of affected animals ends up hole-ridden and resembling a sponge. This perforation of the brain is a common feature shared by all diseases suspected of being caused by infective prions. In sheep, the disease commonly called 'scrapie' occurs. The human form is called nvCJD, which is a new variant of *Creutzfeldt-Jakob Disease,* or *CJD* (pronounced *croits – feld – yak – ob*). The lower-case 'nv' is added to the CJD to indicate that it is a new variant, believed to be caused by prions. Some people refer to it as 'human-form BSE'.

A connection has been made between cattle feed that contained the offal of sheep infected with scrapie and the cattle later developing BSE. Some humans who ate BSE-infected meat have developed nvCJD. Yet the actual infective prions have not yet been isolated and identified in the muscle of any infected beef to prove the connection. This leads researchers to question whether the prions, which clump in the brain, spread out within muscle tissue and are so limited in number that they are nearly impossible to identify. Experiments with mice found tiny pools of infective prions within muscle tissue but only under specific conditions, which may not mirror real life situations.

2 Describe the symptoms of a *coma*.

3 Give a definition for *offal*.

4 Identify the disease described here: *'A disease of the nervous system of sheep that has symptoms of twitching, severe thirst, loss of weight and paralysis.'*

5 Give the full name for the acronym BSE.

6 Give the full name for the acronym CJD.

7 Anyone who ate beef in Britain in the 1980s is not allowed to donate blood in Australia. Give possible reasons for this.

Here are some abbreviations you will encounter when reading about prions:

- **PrPC:** Healthy prions found in cells (*C:* cell)
- **PrP:** Prion protein (*Pr: Pro*tein); indicates the abnormally folded prion linked to BSE
- **PrPSc:** Prions connected with the sheep disease 'scrapie' (*Sc: Sc*rapie)

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BSE-infected brains provide grim visual evidence that the infective agent was present. The deformed prions have been isolated from brain tissue and associated neurons. Prions have not yet been isolated in beef and, although samples look normal, prions may be there somewhere. Only a few misshapen prions might be needed and these can then bind to healthy prions and refold them into disease-causing variants. The biochemical conditions found in brain and neuronal tissue could be important to the onset of infection. Scientists are divided about how disease occurs. The 'Protein-only' theory argues that the misshapen prions are directly responsible for causing disease. Another view is that the misshapen prions get 'outside help' in the form of molecules within the body.

8 What is *a neuron*?

If only a few mutant prions need to be present, this might support the delay in onset. Many disease agents replicate much faster than infective prions. It takes time for them to refold healthy prions. The mutated products would then go on to reshape healthy ones until there were sufficient numbers to produce a disease condition. This takes longer than replicating genetic material – which prions lack. The misshapen form (PrP) could be only a temporary 'middleman' to infection and not, in itself, the direct cause of infection. Experiments were performed in which brain tissue infected with PrPSc (which causes scrapie) was injected into the brains of two groups of healthy mice. One group of mice received the PrPSc combined with the deformed PrP while the other only received pure PrPSc. Where PrP was absent, infection did not occur. Yet, injecting pure PrP into healthy brains also failed to produce the disease. This suggests that a third piece of the puzzle is missing – but what could it be? Enter 'protein X', the suggested middleman. Some scientists believe that a molecule (probably a protein), must somehow get involved in the whole 'bind and reshape' activity. Laboratory experiments in which pure 'scrapie' prions were mixed with healthy prions (but no other molecules were present), gave interesting results.

The mutant PrPSc was able to attach to healthy prions (PrPC) and reshape them but when this happened, the newly deformed (once healthy) prions did not cause infection. They were simply deactivated. Another interesting experiment, using transgenic mice, showed how the different strains might arise based on evolutionary links. *Transgenic* species are those that have genetic material introduced into their genome from another species. A strain of infective prions were extracted from a group of infected hamsters (small rodents) and injected into a group of mice. Although both are related rodent species, infection did not occur. The DNA of the transgenic mice contained some hamster genes and when the same infective prions were injected into this group infection occurred. However, when the prion structure was investigated, it differed slightly to the original material suggesting that the mutant prions are able to 'adapt' and reshape to compensate for genetic differences between related species, thus producing a new strain.

Explain what is meant by the abbreviations:

9 PrPC

10 PrP

11 PrPSc

People are willing to accept that cross-infection happens between sheep and cattle because those two species are reasonably closely linked in evolutionary time. People are less willing to believe that, because of the much longer evolutionary distance between humans and these animals, cross-infection can occur by eating infected meat. This will continue to generate debate, though evidence suggests that, eventually, the proof will be forthcoming.

12 Outline the theories on how prions may cause disease.

13 What is meant by *a transgenic* organism?

Research

14 Since this sheet was written, scientists will have discovered new facts about prions and diseases such as BSE and CJD. Outline any new developments not mentioned here.

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